

CSIS Project/Dissertation Supervisor Directory

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Adetuni, Rose

Key Themes

Digital mental health; Older adults and technology; Health inequalities; Sexual health services; Digital healthcare access.

Sample Projects

- Understanding older adults lived experiences of barriers to digital mental health interventions
- Barriers to Digital Mental Health Interventions in Older Adults
- Digital Mental Health Adoption Among Older Adults in Rural Ireland
- Evaluating Accessibility Barriers in Public Sexual Health Portals
- Deep Reinforcement Learning for Autonomous Vehicle Path Planning in OpenAI Gym
- Analysis of Performance Among Diverse Web Development JavaScript Frameworks.

Albarracin, Juan

Sample Projects

- Attention-based Deep Autoregressive Normal Behaviour Models
- Diffusion-Based Super-Resolution of Wind Turbine SCADA Time Series
- Exploring the capacity of Deep Reconstructive Normal Behaviour Models in Wind Turbine Condition Monitoring

Alrimawi, Faeq

Key Themes

Requirements engineering; Model-driven development; Software testing; Kind computing; Cyber-physical systems and security.

Interested in Supervising

- Student initiated projects: I'm also interested in various software engineering related topics, e.g., investigating requirements for designing and using AI-enabled systems, or developing techniques to enhance the security and forensics of cyber-physical systems.
- Happy to supervise dissertations related to software engineering (SE), especially applying SE techniques (e.g., modelling, automation) to boost kindness of users – you can lookup kind computing for more information. I'm also interested in cybersecurity and digital forensics.
- I'm interested in requirements engineering, model-driven development, automation, and software testing. I'm also interested in kind computing, developing technology that enhance kind interactions between its users. I encourage students to propose their project ideas, preferably around Software Engineering or with a strong Software Engineering component, and we can discuss.

Sample Projects

- Project: Modelling and simulating prosocial behaviour.
Description: this builds on existing meta-model for representing kindness opportunities to create a simulation of potential prosocial behaviour, capturing various factors, including cyber-physical and interactions with the psycho-social. The simulation environment can be a type of a cyber-physical system, e.g., energy-sharing community where people send each other electricity via a shared platform/grid.
- FYP match making system
- Enhancing Fraud Detection Capabilities with Hyper-Automation Technologies
- The Future of Software Testing: An Analysis of Codeless Automation Testing Systems

Buckley, Jim

Key Themes

Software comprehension; Feature location; Architecture recovery; Software metrics; More generally (AI-generated) code analysis. Responsible software engineering, Software engineering pedagogy (in the age of AI?)

Sample Projects

- A Holistic Reference-Breadth (HoReB) grading for academic papers (Category: Research/Prototype Implementation). Focuses on defining a metric to assess the scope of citations in academic papers and implementing a prototype scraping tool.
- C-CEM: Comment Co-Evolution Metric (Category: Research/Software Application Development). Developing a tool to inspect VCSs and evaluate the proportion of code changes that incorporate comment changes.
- Clone Detection QA: Dataset Improvement (Category: Research/Software Application Development). Stress testing clone detection using large datasets and crowdsourcing consensus.

- Clone Detection QA: Quality Approximation Via Comparison (Category: Research/Software). Assessing clone detection recall/precision across result-sets from industrial partners.
- Word Application Domain Match (WADOM) Metric (Category: Research/Some Software Application Development). Deriving and implementing metrics on identifier and comment meaningfulness in domain comprehension for human versus AI-generated code.
- Good Naming Convention (GNC) Metric Evaluation (Category: Software Application Development). Formulating and evaluating GNC metrics against comprehension and feature location literature.
- Quantifying Architecture Consistency (QAC) (Category: Research). Deriving metrics to quantitatively assess the consistency between designed architecture and software implementation.
- Area: Software Comprehension (Research into how programmers understand AI and non-AI generated code; pedagogy).
- Area: Architecture Recovery (Assessing architectural drift and remediation).
- Area: Feature Location (Identifying delocalized functionality in source code).
- Area: Novel Metrics (Comment And Identifier Naming [CAIN] metrics; comparing human vs AI-generated code).
- Area: Research Methods adaptations with AI (Mapping studies and SLRs in Software Engineering).
- Area: Responsible Software Engineering
- Area: Software Engineering pedagogy in the age of AI.

Collins, J.J.

Key Themes

Reinforcement learning; Microservices architecture; Computer vision for nutrition; Evolutionary clustering; Compiler optimization.

Sample Projects

- Deep Q Network versus Trust Region Policy Optimisation for Games (Category: Research / Software Development). Empirical comparison of DQN and TRPO in games (Atari/Connect4/CARLA).
- Compiler Flag Optimisation using Machine Learning (Category: Research / Software Development). Developing a framework with GAs or RL to configure compiler flags (GCC/LLVM-clang).
- Exploring Mechanisms to Resolve Catastrophic Forgetting in Deep Q Networks (DQNs) for the Game of Connect4 (Category: Research / Software Development).
- A Deep Learning Pipeline for Dietetics (Category: Research / Software Development). 3-stage pipeline PoC combining semantic segmentation, food classification, and nutritional database querying.
- Porting from Monolithic to Microservices (Service Cutter, MicroValid framework evaluation).
- Code Smells and Software Metrics for Microservices Architecture.
- Software Clustering using Grammatical Evolution.
- Convolutional Neural Networks for Dietary Analysis (CNNs, Inception V4, ResNet, Food 101 dataset).
- Deep Q Networks for Retro Arcade Games
- Microservice Decomposition Tool
- Deep Reinforcement Learning for Dynamic Traffic Signal Control
- Comparative Analysis of YOLOv8 and Mask R-CNN for Nutritional Portion Estimation

- Mitigation of Catastrophic Forgetting in Deep Q-Networks via Experience Replay Optimization
- Quantitative Evaluation of Architectural Smells in Microservice Decompositions
- Automated Extraction of Microservices from Monoliths Using Service Cutter

Colreavy, Simon

Key Themes

Clinical decision support; Nursing education technologies; Healthcare simulation; EHR usability; Digital health education.

Sample Projects

- Assessing the Relationship Between EHR Cognitive Workload and Clinician Response Accuracy to High-Priority Medication Alerts
- To Explore Student Nurse Perceptions & Attitudes of Virtual Reality in Nurse Education Programmes
- The Impact of Artificial Intelligence on Clinical Outcomes in Minimally Invasive Gynecological Surgery: A Systematic Review and Meta-
- Virtual reality simulations for clinical procedure training
- Usability audits of electronic health records
- Evaluating Cognitive Workload of Alert Fatigue in Intensive Care EHRs
- Virtual Reality Simulation in Undergraduate Nursing Curricula

Crowley, Katie

Key Themes

Digital health; Mental health technologies; Human-computer interaction; Medication safety; Health technology evaluation.

Sample Projects

Digital Health (Category: Software Application Development / Website Development)

- Design and development of digital health applications
- Health information and patient support websites
- Digital tools to support healthcare delivery and engagement

Digital Health (Category: Research)

- Adoption and use of digital health technologies
- Patient and clinician perspectives on health technologies
- Emerging trends in digital health

User-Centered Design & Human-Computer Interaction

- User-centered design of health technologies
- User experience and usability in healthcare applications
- Accessibility and inclusive design for health technologies

Health Technology Evaluation

- Usability evaluation of health applications and websites
- Evaluation of user engagement and technology acceptance
- Assessment of digital health technologies in practice

Dabrowski, Jacek

Key Themes

LLMs for software engineering; Requirements engineering; Mining software repositories; AI developer tools; Tool evaluation and benchmarking.

Sample Projects

- Prompt Me project (<https://prompt-me.lero.ie>). Hands-on experience with Python, LLMs (LLaMA/Mistral), and prompt engineering for Requirements/Software Engineering.
- SimStake: Simulating Stakeholder Personas with LLMs (Category: Engineering/Research).
- SimRE: Requirements Engineer Agent with LLMs (Category: Engineering/Research).
- SimOrch: Orchestration & Shared Memory for Multi-Agent RE Simulations (Category: Engineering/Research).
- Your Own Idea: Exploring LLMs for Software Engineering (Category: Engineering/Research).
- LLMs for Software Engineering (AI assistant for requirements elicitation, test generation, code writing).
- Mining Software Repositories (Extracting actionable insights from GitHub or Play Store).
- Evaluating ML Libraries/Tools (Benchmarking performance, efficiency, and effectiveness).
- Evaluating LLM-Based Multi-Agent Persona Simulations for Requirements Elicitation
- Automated Bug Report Classification and Deduplication Using Transformer Models
- Automated User Story Generation and Refinement Using Instruction-Tuned LLMs
- Benchmarking Code Generation Accuracy and Vulnerability Introduction in AI Coding Assistants

De Lima, Allan

Sample Projects

- Reinforcement Learning for Autonomous Drone Navigation in GPS-Denied Environments
- Multi-Agent Reinforcement Learning for Cooperative Target Tracking

De Wille, Tabea

Key Themes

Localisation and internationalisation; Language technology; Language digitisation; Digital inclusion; Video game development and adaptation.

Interested in Supervising

- Open to supervising projects in international software (localisation/internationalisation), translation technology, the multilingual web, indigenous/minority languages, digital marginalisation/inclusion, language digitization, video games development/adaptation.

Sample Topics

- Game Localisation Pipeline and Translation Management Tool
- Minority Language Keyboard Layout and Input Method Development
- Automated Testing of Internationalisation and Bidirectional Text Rendering in Web Applications
- Evaluating Sustainability and Carbon Metrics in Neural Machine Translation Inference
- Multilingual Accessibility of Digital Health Portals for Refugee and Migrant Populations

Denny, Patrick

Key Themes

Applied machine learning; Computer vision; Autonomous systems; Sports analytics; Explainable AI.

Sample Projects

- Computer vision tracking for athletic velocity analysis
- Edge computing deployments of lightweight vision models
- Embedded Deep Learning for Automotive Camera Perception in Adverse Weather
- Anomaly Detection in High-Volume Industrial Sensor Streams

English, Michael

Key Themes

Software quality; Software maintenance; Feature location; Software metrics; AI in software engineering education.

Interested in Supervising

- Open to supervising projects on Software Engineering topics (software quality, software measurement and metrics, software maintenance, feature location), application of AI in SE, and Computer Science/Software Engineering education.

Sample Projects

- Automated grading and feedback tool for Java assignments
- An evaluation of the adoption of LLM generated code in software development classes
- Evaluating the quality of code generated by LLMs
- The Effect of Class Imbalance Handling on Machine Learning Models for Software Defect Prediction
- Replication and Extension of Baseline IR-Based Feature Location Techniques Using a Fine-Tuned CodeBert Cross Encoder.
- Enhancing Code Smell Detection in Microservices

Healy, Patrick

Key Themes

Combinatorial optimisation; Data visualisation; Sports analytics; Smart travel systems; Explainable AI and algorithms; Operating Systems; Programming Languages; Open Source Software.

Sample Projects

- tab completion over ssh (Category: Software Application Development)
- doodle2graph (Category: Software Application Development)
- Polygon Placement (Category: Software Application Development)
- Sports Analytics I (Category: Software Application Development)
- Sports Analytics II (Category: Software Application Development - hardware sensors and motion modeling)
- Information retrieval: Rank finding for tensors (Category: Research / Software Application Development)
- Drone scheduling / management (Category: Software Application Development)
- EirGrid Cleaner Grid competition (Category: Software Application Development - €2000 prize track)
- Combinatorial Optimisation Problems; Data Visualisation; Problems around smart travel; Explainable AI; DAG generation.
- Graph Drawing and Network Visualisation Engine
- Optimising Sheet Metal Cutting Patterns Using Heuristics
- Combinatorial Optimization for Multi-Modal Urban Journey Planning
- Interpretable Heuristic Search Using Tensor Rank Decomposition
- Exploring new programming languages / paradigms
- Open source software / operating systems.

Hojaji, Fazilat

Key Themes

Data analytics; Machine learning; Open source analytics; Esports analytics; AI-assisted software development.

Interested in Supervising

- software methodologies, project management, data analysis, machine learning applications in video games and healthcare, data governance, and Esports Lab experiments.

Sample Projects

- Predicting Win Chances in League of Legends Using Machine Learning (Category: Predominately Research /Development).
- AI and Machine Learning in Esports: A Systematic Review of Applications and Outcomes (Category: Research).
- Identifying Key Performance Factors in Simulated Racing Using Eye-Tracking and Telemetry Data (Category: Predominately Research /Development).
- KPIs for Measuring Open Source Impact at University (Open Source Program Office - OSPO framework).
- Building an Open Source Impact Dashboard for a University (Automated data pipelines from GitHub/GitLab/repositories).
- AI-Assisted Model-to-Code Generation (Transforming UML/BPMN into working code).

- Automatic Consistency Checking Between Models and Code Using AI.
- Real-Time Telemetry Analysis in Sim Racing for Driver Coaching
- Predictive Modelling of Objective Control in MOBA Esports Matches
- Telemetry and Gaze-Tracking Feature Fusion for Player Skill Assessment in Competitive Gaming
- Automated Traceability Link Recovery Between UML Class Diagrams and Python Code Using Large Language Models

Jose, Abey

Key Themes

Healthcare interoperability; Health information exchange; Digital health implementation; Health innovation ecosystems; Health system sustainability.

Sample Projects

- Topics in Healthcare interoperability
- Multi-System Interactions and Innovation pathways in healthcare
- Barriers to Digital Interoperability in Inclusion Health Services: Implications for Continuity of Care in Older Adults with Complex Needs
- Stakeholder Engagement for the Long-Term Sustainability of Digital Health Implementation

Ju, Andrew

Key Themes

Distributed systems; Algorithms; Machine learning; AI in education; Big data infrastructure.

Sample Projects

- Scheduling via List Coloring (Applying graph theory to resource-constrained scheduling).
- Big Data Teaching Cluster (Kubernetes/Spark cluster deployment for module CS4337).
- AI-Powered Learning Assistant for CS (LLM-driven real-time scaffolding for bi-lingual learners).
- Past MSc AI: Dynamic Resource Allocation in Distributed Computing Using Graph Colouring Heuristics
- Past MSc AI: Adaptive Prompting in Multilingual Educational AI Systems
- Past MSc AI: Performance Characterisation of Big Data Workloads on Kubernetes Clusters

Khalique, Fatima

Key Themes

Privacy-preserving AI; Federated learning; Consent and governance; Explainable AI; Digital epidemiology and synthetic health data.

Sample Projects

- How privacy addressed in AI enabled healthcare applications for disease diagnosis.

- Federated learning based Prediction of Disease.
- Explainable AI for clinical decision support Systems [Allocated: chioma anyaorah]
- Comparative Study of Consent Models Across Health Data Platforms
- Analyze how dynamic patient consent impacts FL utility.
- Use blockchain for transparency in health data AI training.
- Differential Privacy for Healthcare Time-Series Data.
- Privacy and Utility Assessment of Synthetic Health Data Generation.
- Digital Epidemiology Using Non-Traditional Data (mobility, social or environmental data).
- Assessing Consent Models in Multi-Institutional Healthcare Data Sharing
- Privacy-Preserving Machine Learning Architectures for Electronic Health Records

Kshirsagar, Meghana

Key Themes

Healthcare AI; Multi-agent systems; Trustworthy AI; Financial AI and fraud detection; Healthcare operations analytics.

Sample Projects

- Future learning: Generative AI in higher education for diverse student cohorts
- Enhancing the accuracy of large language model's responses with knowledge graphs (GraphRAG)
- Deploying trustworthy AI workflows in high stakes settings (medical diagnosis/fraudulent detection/AI data governance) by incorporating fairness and explainability
- Healthcare Demand Forecasting with predicting length of stay in hospitals for different diseases in Irish Hospitals
- Multi-Agent Systems for Payment Monitoring
- Case study on safety testing of large language models in financial products recommendation
- Intelligent Triage Agents for Streamlining Patient Flow
- Digital Twins for Population-Level Kidney Health Monitoring: A Case Study of the National Kidney Surveillance Programme
- Multi-Agent Systems for behavioural Interventions in Diabetes Prevention
- Tabular synthetic data generation with multi-agent systems (Langchain/Autogen Framework)
- Predictive Modelling of Emergency Department Readmissions
- Explainable Credit Risk Scoring using SHAP and XGBoost
- Graph Retrieval-Augmented Generation (GraphRAG) for Clinical Decision Support
- Fairness-Aware Machine Learning in Automated Loan Underwriting
- Length-of-Stay Prediction in Acute Hospitals Using Multimodal Time-Series Analytics

Le Gear, Andrew

Key Themes

Software maintenance; Architecture recovery; Software product lines; Blockchain applications; AI-assisted learning.

Sample Projects

- Software Maintenance:
 - Automated Architecture Recovery via Feature-Decomposition

- Cross-System Clone Detection for Product Line Migration
 - Real-time Architectural Guardrails (Inverse Reflexion Modelling)
 - Temporal Summarization of Business Processes
 - Multi-Viewpoint Design Recovery Toolset
- Blockchain and AI:
 - Privacy-Preserving Trading via Homomorphic Encryption (HE)
 - Zero-Knowledge (ZK) Strategy Analysis: Trade vs. Owner Privacy
 - Native Android Workflow for Ethereum Transaction Signing
 - AI-Native "Active Learning" for Intelligent Study Systems
 - Optimized LLM Architectures for Constrained Cloud (K8s)
 - Comparative Benchmarking: Railgun vs. Custom ZK-Shielded Pools
- Automated Extraction of Reflexion Models from Dynamic Execution Traces
- Empirical Evaluation of Static vs Dynamic Feature Location in Microservice Systems

Margaria, Tiziana

Key Themes

Neurosymbolic AI; Software development systems; Software verification; Correct-by-design systems; Low-code/no-code engineering; Intelligent workflows and modelling.

Applications in manufacturing, healthcare, bio, history and languages, energy.

Sample Projects

- Neurosymbolic computation and software production for high assurance applications and regulated industries
- DAO-based governance of complex self-managing communities
- Advanced manufacturing in the age of sustainability, e.g., Domain Specific Language editors for applied robotics in ROS
- Advanced workflows for correct by design analytics
- Retargetable recommender systems
- Geoinformation systems and intelligent visualizations for next generation maps
- Document analysis and transcription – advanced workflows for adaptable pipelines
- Metamodelling driven design of Domain Specific Language editors for security, privacy and trust
- Automatic test suite generation for adaptive cyberphysical systems
- Code generation from rich architectural models to Python, Julia and R
- Active Automata Learning for Specification Extraction in Regulated Software Systems
- Model-Driven Code Generation from High-Level Workflows to Distributed Microservices
- Automated Test Suite Generation for Cyber-Physical Systems using LearnLib

Marshall, Mark

Key Themes

- Open to supervise projects in interaction design, physical computing, augmented reality, digital fabrication, technology for museums, digital musical instruments

Sample Projects

- An augmented reality system for artworks in museums
- Wearables for health, sport, disability, etc.
- New musical instruments using sensors and sound synthesis
- Location-based augmented reality

- Tangible interfaces for museums

McElligott, Annette

Sample Projects

- Usability benchmarking of clinical portal interfaces
- Health data governance models in distributed electronic records

Motti Ader, Lilian

Key Themes

User experience design; Accessibility; Health technologies; Educational technologies; Sustainability and digital behaviour.

Sample Projects

- Digital health (Category: Interaction Design / Human-Computer Interaction)
- Educational technologies (Category: Interaction Design / Human-Computer Interaction)
- Environmental sustainability (Category: Interaction Design / Human-Computer Interaction)
- Available to supervise projects with focus on frontend, user interfaces, accessibility (heuristic evaluation, re-design and prototype testing, usability testing, user research):
 - Digital waste and environmental impact
 - Health technologies, ageing and wellness
 - Education and Universal Design for learning
- Accessibility Audit and Interface Redesign for Older Adults
- Mobile Application Supporting Postural Rehabilitation
- Accessibility Testing Automation for Web Applications Using Computer Vision

Nikolov, Nikola

Key Themes

Machine learning; Natural language processing; Network visualisation; Computer vision and OCR; MLOps and model evaluation.

Sample Projects

- Large Network Visualisation and Analysis (Exploring multidimensional scaling and t-SNE vs force-directed layout algorithms; C++).
- Deep Learning for Arabic Manuscript OCR (Deep networks for historical Arabic manuscript digitisation).
- Refactoring a Network Visualisation Application (Porting and refactoring Linux network visualisation system to Windows; C++).
- Force-directed graph layout engines
- Visualizing complex social network clustering in WebGL
- Deep Convolutional and Recurrent Neural Networks for Historical Arabic Script Recognition
- Dimensionality Reduction Techniques for Multi-Scale Network Visualisation
- Refactoring, Modernization and Porting of Linux Network Visualisation Frameworks to Windows

Noll, John

Sample Projects

- Open to supervise projects in software engineering, open source, and developer collaboration
- Empirical analysis of developer collaboration patterns on GitHub
- Mining open-source code repositories for testing practices

O'Connor, Alison

Key Themes

Digital health systems; Healthcare scheduling; Clinical workflow optimisation; Health data analytics; LLMs in healthcare.

Sample Projects

- Large Language Models for Athletic Nutrition using original anonymous transcriptions
- Large Language Models for Athletic Nutrition using clean and anonymous transcriptions
- Machine Learning in Healthcare
- Investigating bias in algorithmic transcriptions
- Developing an auditable patient scheduling system
- Developing an interface/dashboard application for surgery scheduling
- Scoping agentic LLMs for specific purposes in healthcare
- Developing interactive HTML flowcharts to support student ethics applications within UL and external (HSE) pathways
- Enhancing cross-agency digital communication in medical scheduling
- Exploratory Data Analysis for Theatre Scheduling
- Automated Data Collection and Visualisation for Athletic Performance
- Interactive Tools for Ethics Review Pathways
- Thematic Analysis of Qualitative Health Interviews Using Fine-Tuned LLMs
- Rashomon Set Exploration for Interpretable Clinical Decision Support Models
- Developing Auditable Patient Scheduling Systems
- Interactive Flowcharts for Research Ethics Navigation in Irish Healthcare

O'Connor, Neil

Key Themes

Music technology; Electronic music composition; Sound engineering; Audio production; Digital music systems.

Sample Projects

- Electronic Music Composition: Past and Future (Category: Composition / Music Technology). Examination of music technology in UL EMS studios.
- Electronic Music Composition: Modular Synthesis (Category: Composition / Music Technology). Portfolio of electronic music works (approx. 3 pieces c.10 mins each).
- Music Production Composition Portfolio (Category: Sound Engineering / Music Technology). Portfolio development in DAW or UL UG studio environment.
- Multichannel Electronic Music Composition and Spatialization
- Hardware Modular Synthesis and Signal Processing Portfolio
- Sound Design for Immersive Environments

O'Murchú, Nóra

Key Themes

Interaction design; Digital wellbeing; Media ecologies; Environmental sustainability; Critical technology studies.

Sample Projects

- Resisting the attention economy (Category: Interaction Design). User-centered design to reduce screen time and resist online manipulative tactics.
- Interaction Design for Navigating the Politics of Images in Online Spaces (Category: Interaction Design). Interactive platform to critically engage with visual politics.
- Reducing the Environmental Impact of AI in Interaction Design (Category: Interaction Design). Energy-efficient AI interface prototype.
- Critical Design Interventions for Digital Wellbeing
- Interactive Systems for Navigating Algorithmic Bias
- Visualizing the Material and Carbon Footprint of AI Infrastructure

Parmar, Robin (not available AY2026/27)

Key Themes

Audiovisual systems; Generative art; Video installation; Media theory; Creative coding and visualisation.

Sample Projects

- Video system for collaborative audiovisual performance (Category: Software Application Development)
- Retro computer visuals (Category: Video, Software Application Development)
- Lumia as light sculpturing (Category: Video, Installation)
- Music video as experimental form (Category: Video)
- Generative art as biological process (Category: Video, Installation, Software Application Development)
- The uncanny and genre film (Category: Video)
- The new public: video as pervasive media (Category: Video, Installation, Software Application Development)
- Representing the Anthropocene using data transduction (Category: Video, Software Application Development)
- Generative Audiovisual Performance Environment in Processing
- Kinetic Light Sculptures and Lumia Projections
- Experimental Music Video Production with Custom Analog Glitch Pipelines

Razzaq, Abdul

Key Themes

AI for software engineering; Source code analysis; Developer experience; Software metrics; NLP and LLM applications.

Sample Projects

- Cross-Lingual Text Generation and Translation Using GANs
- Context-Aware Bug Localization Using LLMs
- eBPF-Driven Power Efficiency Analyzer for LLM Workflows

- LLM Training Workflow Observability and Cost Optimization Convention
- Dynamic LLM Workflow Optimization Using Observability-Driven Reinforcement Learning
- Domain and Range of Developer Experience
- Automating Architecture Drift Detection
- Feature Location Using AI-Assisted Techniques
- Measuring and Enhancing Developer Experience (DX) in Collaborative Teams
- Mapping Study on Software Comprehension Tools
- Multimodal Software Metrics
- Framework for Evaluating and Improving Software Sustainability
- Hardware Performance Monitoring using EBPF
- An Automated Tool for Architecture Drift Prediction in Software Systems Using Machine Learning
- Developing a Pipeline to Fine-tune Small Open-Source LLMs for Natural Language to Code Generation in Java
- Enhancing Software Architecture Drift and Erosion Detection Using Natural Language Processing
- Generating Context-Aware Inline Code Comments from Commit Messages and Diff Analysis
- LLM-Powered Chatbot with Continuous Feedback for Customer Support in Telecommunications
- Multi-Perspective Code Smell Detection Using Ensemble Machine Learning Models
- Optimizing Large Language Model Workflows using Reinforcement Learning
- The Empirical Study to Evaluate the Capabilities of Generative AI for Detecting Code Clones in Java
- Automated Requirements Categorisation and Prioritization from User Feedback Using Machine Learning
- Evaluating the Impact of Code Obfuscation on Neural Code Generation Models
- Evaluating the Maintainability of AI-Generated vs Human-Written Microservices Code
- Performance Evaluation of Large Language Models in Automated Python Code Generation
- Source Code Transformation Using Large Language Models: A Comparative Evaluation

Ryan, Alan

Key Themes

User-centred design; Sustainability; Public-sector digital services; Design Research; Critical and Speculative Design.

Sample Projects

- A Critical/Speculative Design Approach to _____. Exploring societal issues and creating a discursive space.
- A User-Centered Design Approach to _____. Iterative design and a working prototype.
- Critical Design Domains: Social Media Addiction, Artificial Intelligence, etc.
- User-Centered Design Domains: Utility-driven Interfaces (Video, Music, Administration), Communication Platforms, Neurodivergence, all with iterative design and a working prototype.
- Redesigning _____ with skeuomorphism in mind: physical realism vs. digital performance (Speculative Design or User-Centered Design).
- Speculative Design for Public Health to encourage Healthier Habits
- User-Centered Design for Public Health to encourage Healthier Habits

- User-Centred Design of a Digital Public Transport Navigation Interface
- Investigating the User Experience in a Smart Home
- Exploring the Materiality of Data: Physical Data Visualisations of Energy Consumption, Waste Products, the Circular Economy
- Limerick Flood Prevention: A public point-of-single-contact flood prediction website.
- Limerick Flood Prevention: A physical artefact that increases awareness/understanding of the relevant issues.

Ryan, Conor

Key Themes

Evolutionary computation; Grammatical Evolution; Automatic program generation; Machine learning for games; Explainable AI.

Sample Projects

- EvoGame: Evolving Game Strategies Using GE
- EvoArt: Evolving Digital Artwork with GRAPE
- EvoWeather: Evolving Weather Forecasting Models
- EvoRace: Evolving Autonomous Racing Car Controller - TORCS simulator
- Project 5: AI Scriptwriter for Short Films (Category: Predominately Research - fine-tuning small LLM on screenplays).
- Project 6: Synthetic Cancer Cell Image Generation (Category: Predominately Research - GANs using TCIA).
- Project 7: Synthetic Multimodal Cancer Data Generation (Category: Predominately Research - conditional GANs on TCGA).
- Project 8: A Bare Metal Implementation of a C++ Program on a Raspberry Pi (Category: Predominately Research).
- Potential MSc Projects: Automatic program generation; Machine Learning for game playing; LLM-related projects; Automatic programming of simulated robots with Grammatical Evolution; Explainable/Fair AI.
- Past FYP: Automatic Generation of Python Functions Using Grammatical Evolution
- Past FYP: Evolutionary Art Generation via Interactive Genetic Algorithms
- Past FYP: Evolving Heuristic Strategies for Board Games Using GRAPE
- Past FYP: Bare-Metal Real-Time Audio Processing on ARM Microcontrollers
- Past MSc AI: Evolving Interpretable Machine Learning Models Using Grammatical Evolution
- Past MSc AI: Conditional Generative Adversarial Networks for Multimodal Cancer Imaging Synthesis
- Past MSc AI: Grammatical Evolution for Automated Deep Learning Architecture Search

Saay, Salim

Key Themes

Software architecture; Model-driven engineering; Blockchain; Event-driven systems; Organisational collaboration systems.

Sample Projects

- Online Voting System (Category: Requirements Engineering, Software Engineering - blockchain-based authentication).

- Integration of AI and Blockchain for collaboration of organizations (Category: Software Architecture / Software Development - GDPR and legal compliance automation).
- Interface development of Kafka Broker integrated with Fabric systems
- Integration of AI with Kafka Broker and Blockchain
- Deployment of DAOs in software system collaborations
- Educational Gap Analysis Systems
- Secure Distributed Voting Architecture Using Hyperledger Fabric
- Model-Driven Engineering of Collaborative Web Applications
- Integration of Apache Kafka with Enterprise Relational Databases
- Blockchain-Driven Smart Contracts for Automated GDPR Compliance in Cross-Organizational Workflows
- Event-Driven Architecture Optimization in Kafka-Based Microservices

Simpson, Jürgen

Key Themes

Electronic music systems; Algorithmic composition; Installation art; Soundscape research; Interactive performance technologies.

Sample Projects

- Developing a historically informed Live Electronics Music interface and performance practice (Category: Composition, Software Application Development).
- Microtonal Music Software and Composition (Category: Composition, Software Application Development).
- Translating Landscape in Video Art / Installation Art Settings (Category: Film, Installation).
- Silent Symphonies: Exploring Biophony Degradation through electronic music (Category: Composition, Installation).
- Algorithmic Horizons: Expanding Composition through Contemporary Software (Category: Composition, Performance Installation - SuperCollider / MaxMSP).
- Spatial Audio Composition in Multi-Channel Surround Systems
- Generative Algorithmic Synthesizers in MaxMSP
- Interactive Sensor-Driven Audio Installations
- Environmental Soundscape Recording and Granular Re-synthesis

Storni, Cristiano

Key Themes

Human-centred design; Participatory design; Sustainability; Inclusion; Social innovation; self-care.

Sample Projects

- Co-designing self-management technologies for chronic illness
- Designing accessible interfaces for older adults in smart environments
- Digital tools for participatory community governance

Torre, Giuseppe

Sample Projects

- Live coding audiovisual performance environments
- Open-source digital audio workstation plugin development

Vassev, Emil

Key Themes

Autonomic systems; format modelling; software engineering

Sample Projects

- Formal Specification of Autonomic Software Systems
- Policy-Based Self-Healing Architectures in Distributed Systems

Wade, Gavin

Key Themes

Game development; Serious games; XR/VR; Games industry studies; Technical game systems (incl. game engine design/development, game artwork techniques and tools).

Sample Projects

- Game Archaeology (Category: Game Development - Programming). Recreating undocumented legacy 32-bit iOS Objective-C game in Unity and C#.
- Porting Legacy Graphics Code to Modern Game Engines
- Game Level Design Theory/Experiments
- Game Projects for a Real-World Client
- Serious Games for Workplace Safety Induction in Virtual Reality
- Comparative Evaluation of Physics Simulation Accuracy in Unity vs Unreal Engine

Ward, Nicholas

Key Themes

Digital media; music technology; interactive media. Tangible interaction, Sound art installations. Wearables. Physical devices (combine electronics and woodwork and digital fabrication in the makerspace to built things).

Sample Projects

- Sonification/Visualisation of data as the basis for art installations, science communication or activism. E.g. developing an artwork that draws from data to communicate climate change.
- Haptic audio-tactile feedback controllers (exploring vibration as a means of giving feedback when interacting with a digital device).
- Projects using touchdesigner, max/msp,PD, Arduino, esp32.
- Projects that lean towards physical devices that you make yourself that combine some sensors and microcontrollers e.g. novel games with bespoke novel controllers,
- Live music or video performance systems using sensors and synthesisers. Interactive physical interfaces for music performance
- New music synthesiser designs based on simple analogue oscillators combined with arduino style microcontrollers.
- Critical Design and Speculative Design: projects where we design and construct devices whose role is to provoke people to think more deeply about a topic (often a topic related to digital culture e.g. surveillance, social media effects, smartphone usage etc).
- Drawing machines: using code and stepper motors to create a machine that draws with a pencil on a big wall. What does it draw? Why does it draw? Is it a creative machine?